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Secure Distribution Venture Pvt. Ltd.

What Is EPC Funding ?

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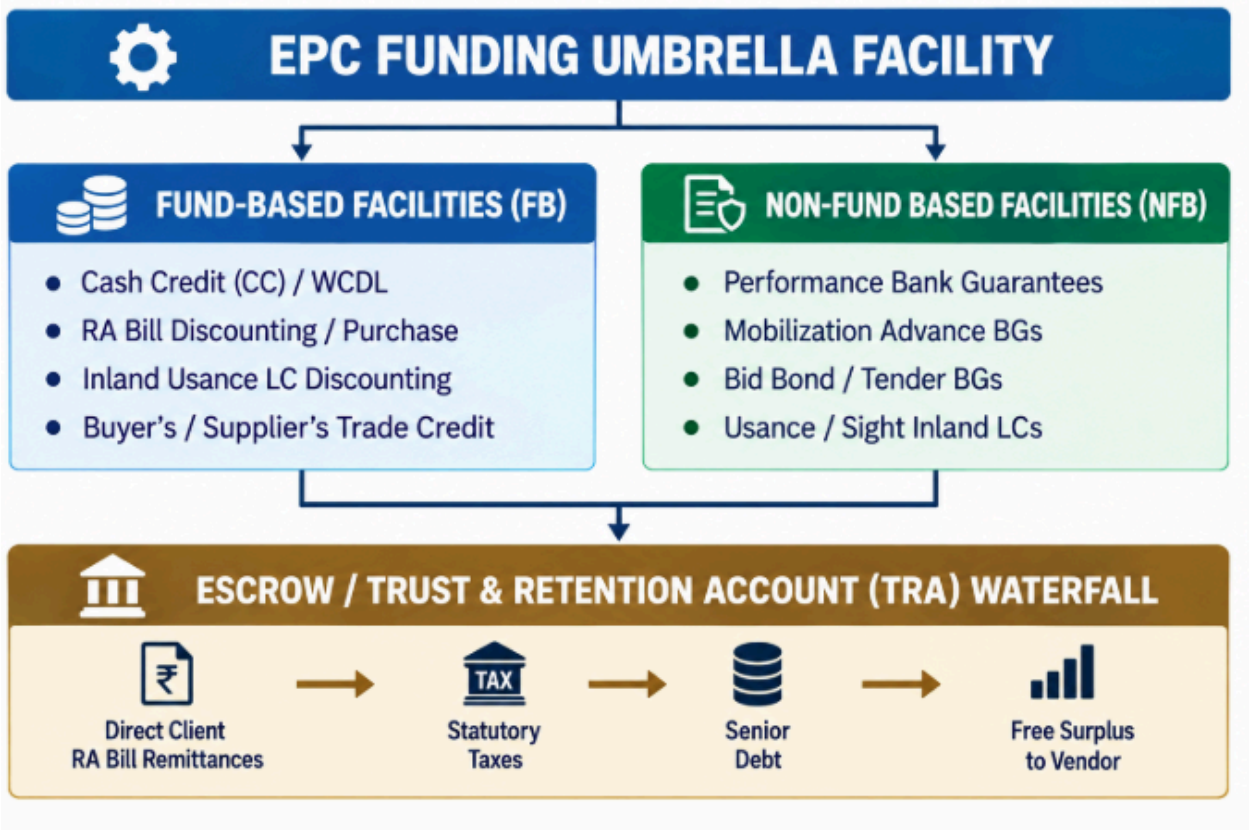
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Executive Summary & Curriculum Architecture

This White Paper Manual serves as a definitive institutional guide to **Engineering, Procurement, and Construction (EPC) Funding** within the Indian financial, banking, and regulatory ecosystem. Designed specifically for corporate treasurers, CFOs, commercial credit underwriters, project developers, and finance professionals, this document provides an exhaustive, mathematically rigorous, and legally grounded exploration of how turnkey contractors secure, structure, and manage multi-tiered credit facilities.



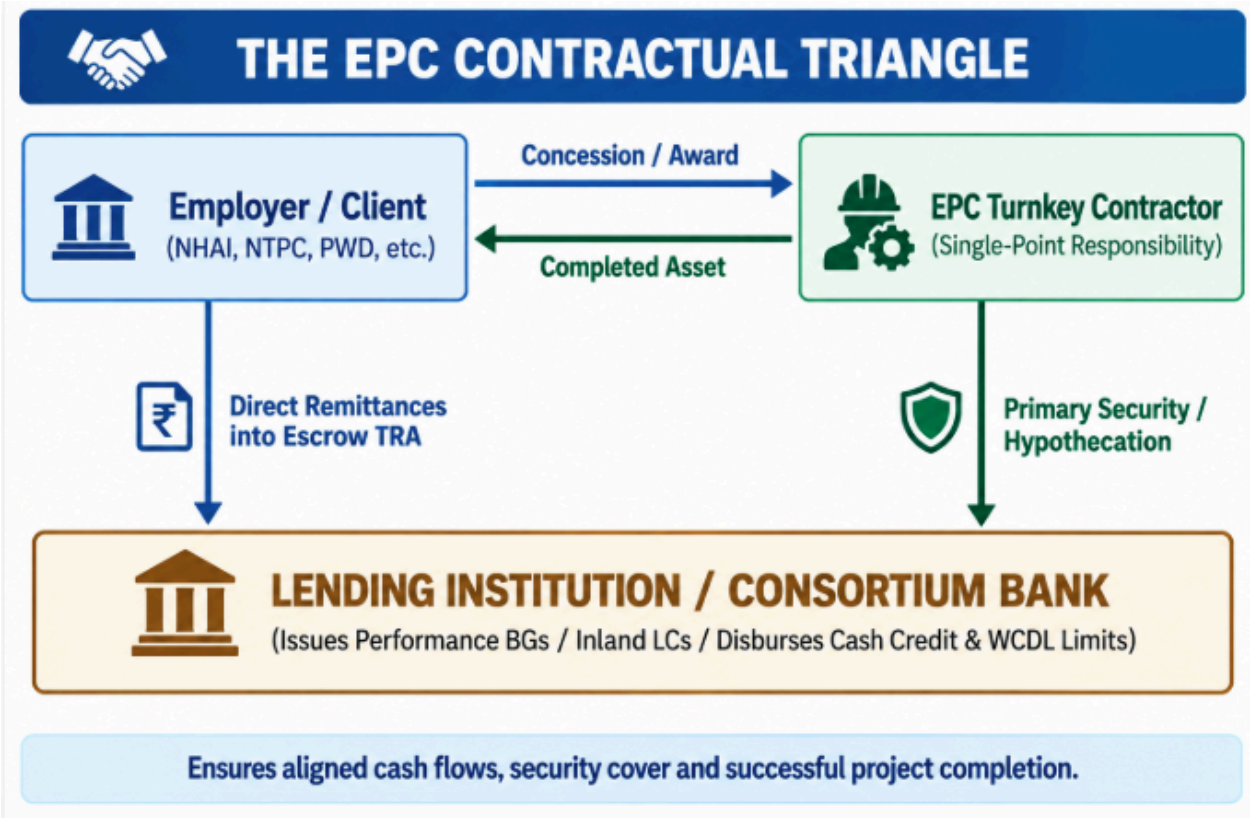
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Module 1: Product Overview & Theoretical Foundations

1.1 Definition & Fundamental Mechanics of EPC Funding

Engineering, Procurement, and Construction (EPC) Funding represents a specialized domain of corporate credit and structured trade finance designed to fund the cash flow, working capital, and guarantee requirements of turnkey contractors. In an EPC contract, the contractor assumes single-point responsibility for designing (Engineering), sourcing materials and equipment (Procurement), and building (Construction) a physical infrastructure or industrial asset—ranging from national highways, railway corridors, power transmission lines, and water treatment networks to renewable energy parks and heavy factory complexes.

Unlike standard commercial term loans that finance asset acquisition on a borrower's balance sheet, EPC Funding is an **order-book and cash-flow-driven credit umbrella**. It bridges the massive liquidity gap created between the contractor's immediate cash outflows for raw material procurement, sub-contractor payments, and equipment deployment, and the employer's delayed cash inflows disbursed via certified Running Account (RA) bills.



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1.2 Contracting Models: LSTK, Item Rate, Cost-Plus, and EPCM

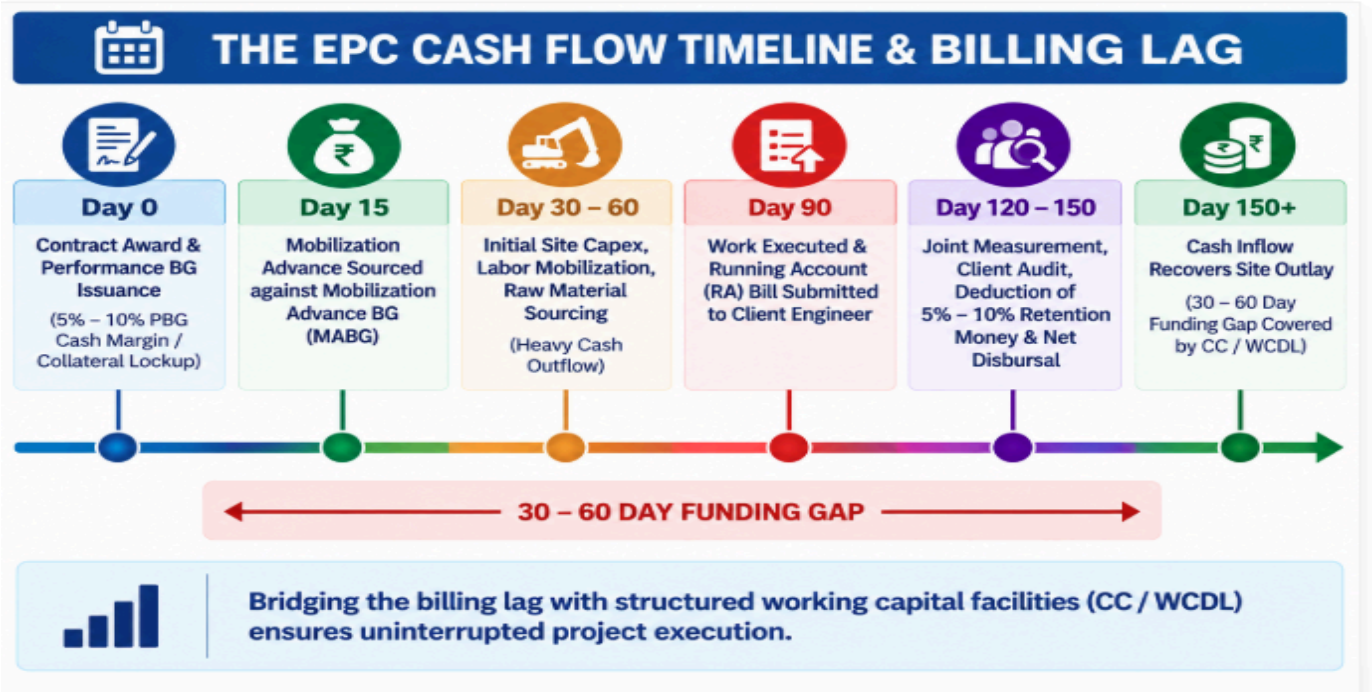
To underwrite EPC Funding accurately, financial underwriters evaluate the underlying contractual structure, as risk distribution varies significantly across four primary contracting models:

1. **Lump-Sum Turnkey (LSTK) Contracts:** The contractor agrees to complete the entire project for a fixed, predetermined total price. All cost overruns resulting from raw material inflation, engineering revisions, or supply chain bottlenecks are borne entirely by the contractor. LSTK contracts require higher contingency cushions and strict price-variation clauses (PVC) before lenders approve fund-based working capital limits.
2. **Item Rate / Unit Price Contracts:** The contractor quotes unit rates for specific bill-of-quantities (BOQ) items (e.g., per cubic meter of concrete laid or per metric ton of structural steel erected). Quantities are measured jointly on-site via Running Account (RA) bills. Price escalation clauses are tied to official wholesale price indices (WPI), offering lenders moderate margin protection.
3. **Cost-Plus Fee Contracts:** The client reimburses the contractor for all actual project expenses plus an agreed profit margin percentage or fixed fee. While commercial risk is minimal, lenders monitor cost audit trails to prevent dispute-led payment withholding by the employer.
4. **Engineering, Procurement, and Construction Management (EPCM):** The contractor acts purely as a professional management consultant, providing engineering designs and overseeing procurement/construction without executing the physical works or carrying raw material inventory risks. Funding requirements in EPCM are restricted to low-margin office working capital lines rather than heavy non-fund-based guarantee facilities.

1.3 The EPC Working Capital Mismatch Paradox

Turnkey contractors operate under a severe, structural cash flow bottleneck known as the **EPC Working Capital Mismatch Paradox**. Unlike manufacturing or retail businesses with rapid inventory turnover, EPC execution is characterized by asymmetric cash cycles:

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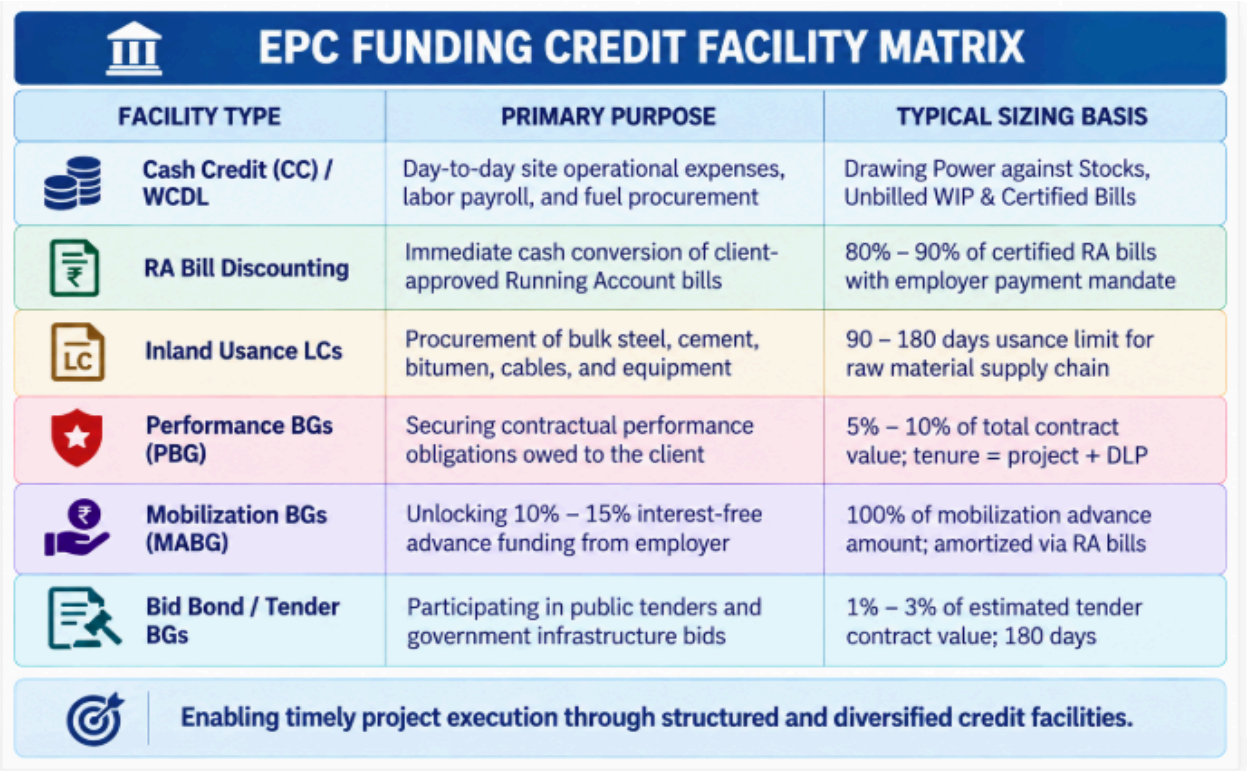
Key contractual drivers of the liquidity paradox include: * **Site Mobilization Outlays:** Setting up batching plants, labor camps, crushing units, and mobilizing heavy equipment fleet requires immediate cash before Mobilization Advance BGs are accepted. * **RA Bill Certification Lags:** Joint measurement, site engineer verification, quality testing, and bureaucratic approvals result in a 60-to-120 day delay between physical work execution and final cash receipt. * **Contractual Retention Money Locks:** Employers routinely deduct 5% to 10% from every certified RA bill as “Retention Money” to guarantee defect rectification. This cash remains locked for the entire construction period plus the Defect Liability Period (DLP) of 12 to 24 months, consuming the contractor’s net profit margin. * **Guarantee Margin Lockups:** Banks require 10% to 25% cash margin in the form of Fixed Deposits (FDs) to issue Bank Guarantees (Performance BGs, Bid Bonds, MABGs), restricting promoter liquid equity.

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Module 2: Structural Architecture & Facilities Breakdown

2.1 The Dual Fund-Based & Non-Fund Based Credit Architecture

EPC Funding operates as a comprehensive credit umbrella combining **Fund-Based (FB)** liquidity facilities with **Non-Fund Based (NFB)** trade guarantees:



EPC FUNDING CREDIT FACILITY MATRIX

FACILITY TYPE	PRIMARY PURPOSE	TYPICAL SIZING BASIS
Cash Credit (CC) / WCDL	Day-to-day site operational expenses, labor payroll, and fuel procurement	Drawing Power against Stocks, Unbilled WIP & Certified Bills
RA Bill Discounting	Immediate cash conversion of client-approved Running Account bills	80% – 90% of certified RA bills with employer payment mandate
Inland Usance LCs	Procurement of bulk steel, cement, bitumen, cables, and equipment	90 – 180 days usance limit for raw material supply chain
Performance BGs (PBG)	Securing contractual performance obligations owed to the client	5% – 10% of total contract value; tenure = project + DLP
Mobilization BGs (MABG)	Unlocking 10% – 15% interest-free advance funding from employer	100% of mobilization advance amount; amortized via RA bills
Bid Bond / Tender BGs	Participating in public tenders and government infrastructure bids	1% – 3% of estimated tender contract value; 180 days

Enabling timely project execution through structured and diversified credit facilities.

2.2 Detailed Mechanics of Fund-Based Facilities

1. **Cash Credit (CC) / Working Capital Demand Loan (WCDL):** Revolving debt facilities accessed by the contractor to fund site operations. Drawing Power (DP) is calculated monthly by applying a standard haircut (typically 25% on raw material inventory, 25%–30% on unbilled Work-in-Progress, and 15%–20% on certified receivables under 90 days).
2. **Running Account (RA) Bill Discounting / Purchase:** A specialized short-term facility where the bank purchases or discounts client-certified RA bills at a fixed spread over EBLR/MCLR. Funds are disbursed directly to the contractor's operational account, and client remittances are routed directly to the bank's escrow account to extinguish the loan.
3. **Inland Usance Letters of Credit (LCs):** Irrevocable trade instruments issued by the contractor's bank to raw material suppliers (e.g., steel and cement OEMs). Suppliers receive immediate payment by discounting the LC at their bank, while the EPC

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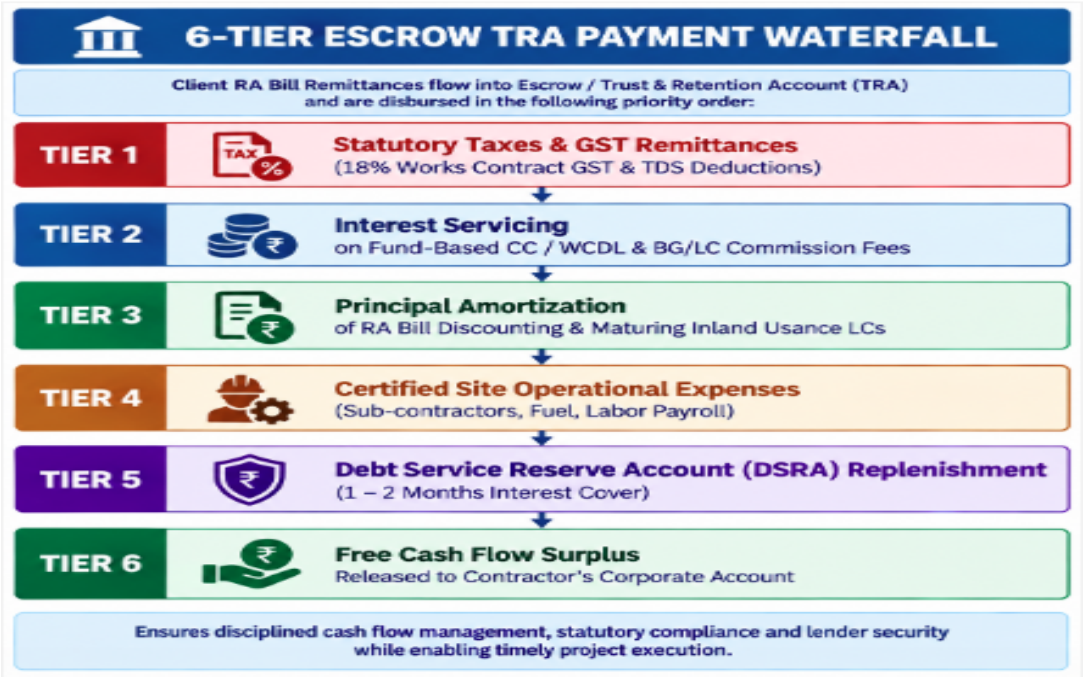
contractor gains 90 to 180 days of usance credit to align raw material costs with project milestone billings.

2.3 Detailed Mechanics of Non-Fund Based Guarantee Facilities

1. **Performance Bank Guarantees (PBG):** Issued to the employer (e.g., NHAI, NTPC) for 5% to 10% of the total contract value. If the contractor fails to perform per contractual specifications, the employer can invoke the PBG on demand. Banks underwrite PBGs by evaluating contractor technical capability, historical track record, and current order book execution capacity.
2. **Mobilization Advance Bank Guarantees (MABG):** Employers grant an interest-free or concessional advance (typically 10% to 15% of contract value) to assist contractors with site setup. To receive this advance, the contractor submits an equivalent MABG to the employer. The advance is recovered proportionally via deductions from subsequent RA bills, leading to a systematic reduction in MABG exposure over time.
3. **Retention Money Bank Guarantees (RBG):** Contractors submit RBGs to employers to release the 5% to 10% retention money locked from RA bills. This converts locked cash receivables into immediate liquid working capital, improving contractor cash velocity.

2.4 Escrow & Trust & Retention Account (TRA) Ring-Fencing

To insulate lenders from cash diversion across multiple project sites, EPC Funding mandates a **Trust and Retention Account (TRA) Escrow Structure**. The employer executes a direct payment mandate agreeing to deposit all certified RA bill remittances into a dedicated bank-controlled TRA Escrow Account.



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Module 3: Comparative Capital Architecture Matrix

COMPARATIVE CAPITAL ARCHITECTURE MATRIX

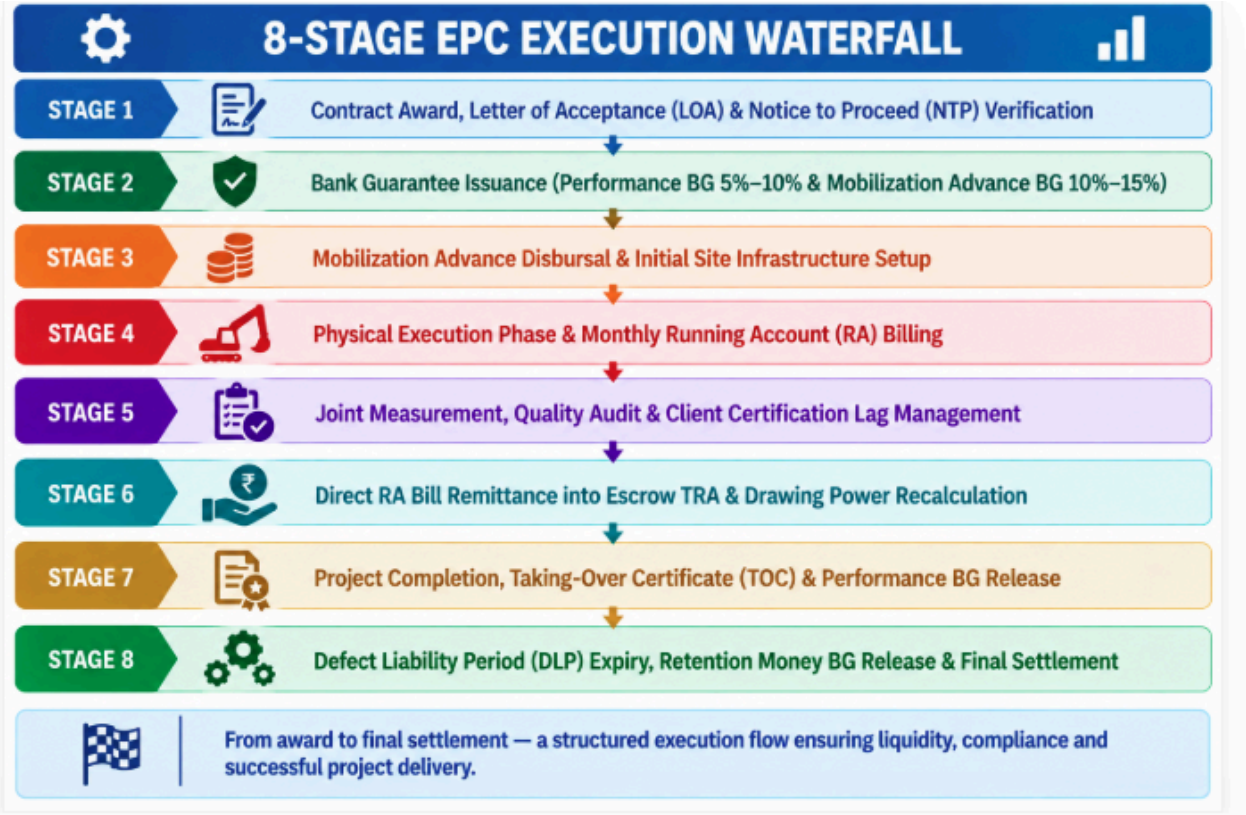
PARAMETER	EPC FUNDING UMBRELLA	PROJECT FINANCE SPV DEBT	GENERAL CORPORATE TERM LOAN	INVOICE FACTORING / TReDS
Primary Target Asset / Purpose	Order-book execution, site mobilization, BGs & LCs	Greenfield asset creation (Solar, Highway, Port SPV)	General corporate expansion or debt refinancing	Certified short-term trade receivables
Primary Borrower Entity	Turnkey EPC Contractor (Operating Entity)	Bankruptcy-Remote SPV Subsidiary	Operating Parent Company	Vendor / Supplier MSME
Facilities Composition	FB (CC/WCDL/Bill Disc.) + NFB (BGs, Usance LCs)	Long-Term Fund-Based Project Term Loan	Pure Fund-Based Term Loan	Pure Fund-Based Receivable Discounting Line
Drawing Power / Haircut Margin	15% – 25% on Inventory/WIP; 10% – 25% Cash Margin for BGs	N/A (Debt-to-Equity LTV 70:30 to 80:20)	N/A (Fixed asset coverage ratio FACR >= 1.25x)	10% – 15% haircut on invoice face value
Indicative Benchmark Pricing	FB: EBLR/MCLR + 1.50% – 3.50% NFB Fee: 0.75% – 2.25% p.a.	EBLR/MCLR + 2.00% – 4.00% p.a.	EBLR/MCLR + 2.50% – 4.50% p.a.	TReDS Bidding: 7.50% – 9.50% p.a. annualized rate
Indicative Tenure / Horizon	12-Month Revolving Line (BGs: 1 – 5 Years per BG)	15 – 25 Years (Matching Concession / PPA Life)	5 – 8 Years Door-to-Door	30 – 180 Days Per Bill
Primary Security Hierarchy	Exclusive First Charge on WIP, Stocks & RA Bills	Exclusive First Charge on SPV Assets & PPA Cash Flow	First Charge on Land, Factory & Machinery	Direct Assignment of Trade Receivables
Collateral Coverage Requirement	30% – 50% Real Estate / Industrial Land Mortgages	Pledge of 51% – 100% SPV Equity Shares	100% – 150% Fixed Asset Coverage	Unsecured / Non-Recourse (Anchor Buyer Credit)
Core Coverage Ratio Benchmark	DSCR >= 1.35x; ICR >= 2.0x; Current Ratio >= 1.33:1	DSCR >= 1.35x; LLCR >= 1.40x	DSCR >= 1.35x; Debt-to-EBITDA <= 3.5x	N/A (Anchor Buyer Credit Rating Benchmark)
Escrow TRA Control Level	Mandatory Project Escrow TRA Account	Mandatory Ring-Fenced SPV Escrow Waterfall	Optional Cash Flow Escrow	Direct Platform Payment Settlement
Section 36(1)(iii) Tax Shield	Fully Tax Deductible (FB Interest + NFB Fees)	Fully Tax Deductible (SPV Debt Interest)	Fully Tax Deductible (Corporate Debt Interest)	Fully Tax Deductible (Discounting Cost)

Different structures. Different purposes. A common goal — enabling growth with the right capital, structure and risk mitigation.

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Module 4: Operational Lifecycle & 8-Stage Execution Waterfall

The operational lifecycle of EPC Funding follows an 8-stage execution waterfall, aligning credit disbursals with contractual milestones and statutory legal compliance:



4.1 Stage-by-Stage Breakdown

- **Stage 1: Contract Award & Due Diligence:** Upon winning a public or private tender, the contractor receives the Letter of Acceptance (LOA). The lender reviews the LOA, Concession Agreement, Price Variation Clause (PVC), escalation caps, and client creditworthiness.
- **Stage 2: Bank Guarantee Issuance:** The lender issues the mandatory Performance Bank Guarantee (PBG) and Mobilization Advance BG (MABG) to the employer. The contractor deposits the required cash margin (10%–20% FD) and executes counter-indemnities.
- **Stage 3: Mobilization Advance Disbursal:** The employer disburses 10% to 15% interest-free mobilization advance directly into the project TRA Escrow Account upon receiving the MABG. Funds are utilized for site clearing, batching plant setup, and labor camps.

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- **Stage 4: Execution Phase & RA Billing:** As civil execution progresses, the contractor submits monthly Running Account (RA) bills detailing work completed per the bill-of-quantities (BOQ).
- **Stage 5: Joint Measurement & Client Certification:** The client's engineer inspects the site, verifies quality standards, and certifies the RA bill—typically deducting 5%–10% retention money and a pro-rata mobilization advance recovery.
- **Stage 6: Escrow Remittance & Drawing Power Recalculation:** Certified net funds are remitted directly by the employer into the bank's Escrow TRA. The lender recalculates monthly Drawing Power (DP), reinstating the contractor's cash credit line.
- **Stage 7: Project Completion & TOC Issuance:** Upon physical completion, the employer issues the Taking-Over Certificate (TOC). The MABG is fully released, and the PBG exposure is reduced or converted into a Maintenance/DLP BG.
- **Stage 8: Defect Liability Period Expiry & Final Settlement:** Upon expiry of the 12-to-24 month Defect Liability Period, the employer releases all remaining Retention Money or cancels the Retention BG, fully extinguishing the facility.

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Module 5: Common Eligibility Criteria & Underwriting Framework

5.1 Working Capital Sizing Methodologies

Indian commercial banks and financial institutions utilize three primary regulatory methodologies to underwrite EPC fund-based and non-fund-based credit limits:

1. Nayak Committee Turnover Method (For Limits <= ₹5.00 Crores)

For small-scale EPC contractors and sub-contractors requiring total credit limits up to ₹5.00 Crores, working capital is sized as a flat percentage of projected annual turnover: * **Minimum Working Capital Assessment:** 25% of projected annual gross revenue. * **Bank Credit Contribution (FB Limit):** 20% of projected annual gross revenue. * **Promoter Margin Contribution:** 5% of projected annual gross revenue.

2. Tandon Committee Second Method of Lending (MPBF) (For Limits > ₹5.00 Crores)

For medium and large EPC contractors requiring credit limits exceeding ₹5.00 Crores, lenders apply the Maximum Permissible Bank Finance (MPBF) formula:

$$Total\ Current\ Assets\ (TCA) = Inventory/WIP + Certified\ Receivables + Other\ Current\ Assets$$

$$Other\ Current\ Liabilities\ (OCL) = Trade\ Creditors + Statutory\ Dues + Advance\ Payments$$

$$Working\ Capital\ Gap\ (WCG) = TCA - OCL$$

$$Minimum\ Stipulated\ Net\ Working\ Capital\ (NWC) = 0.25 \times TCA$$

$$Maximum\ Permissible\ Bank\ Finance\ (MPBF) = WCG - (0.25 \times TCA) = 0.75 \times TCA - OCL$$

Under the Tandon Second Method, the contractor must maintain a **Current Ratio (TCA / TCL) of at least 1.33:1**, ensuring that 25% of gross current assets are funded via long-term promoter equity or retained earnings.

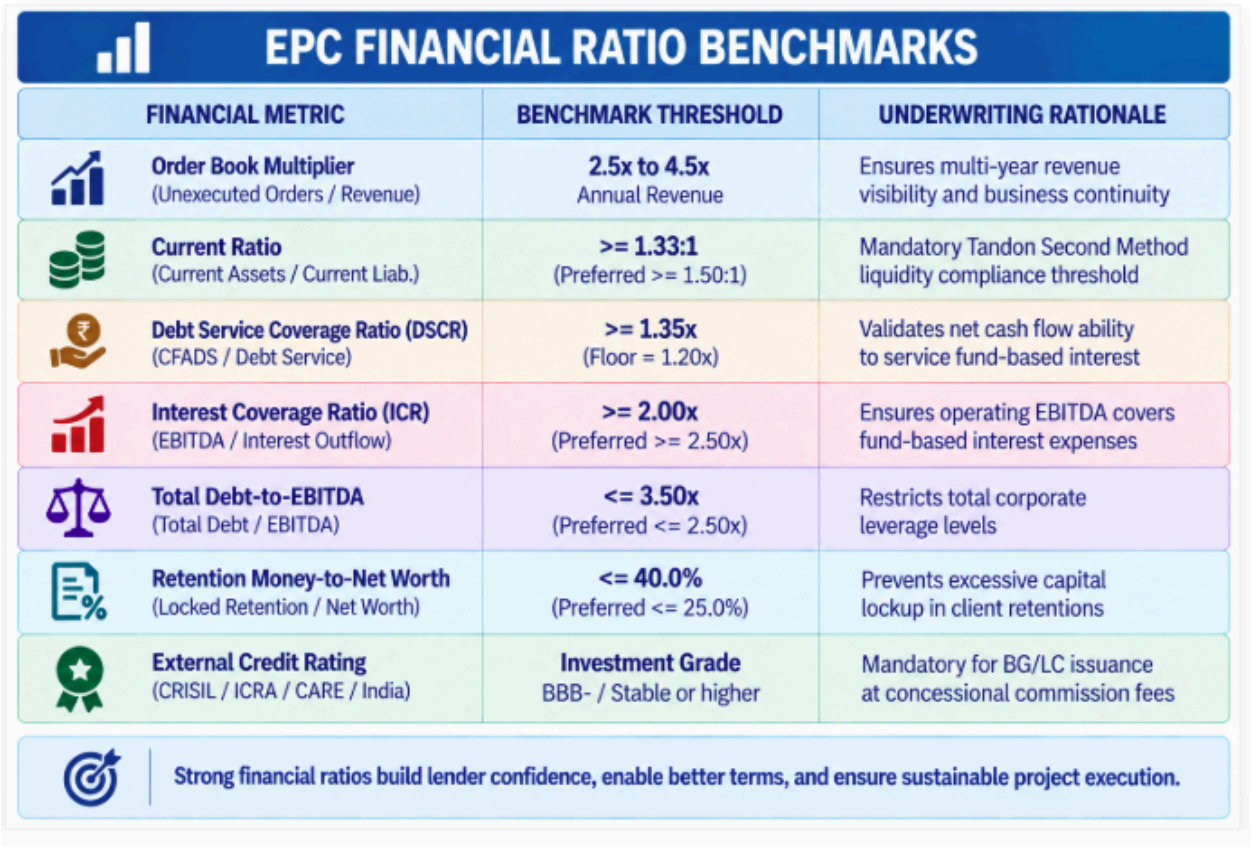
3. Cash Budget Method (For Multi-Year Infrastructure Contracts)

For complex, multi-year EPC contracts (e.g., NHAI HAM roads, metro corridors, power plants), banks require month-by-month Cash Flow Projections over a 12-to-36 month horizon. Credit limits are sized to cover the **peak cumulative cash deficit** identified in the project's construction cash budget.

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5.2 Core Financial Ratio Benchmarks

Underwriters evaluate specific quantitative financial ratios before approving EPC umbrella lines:

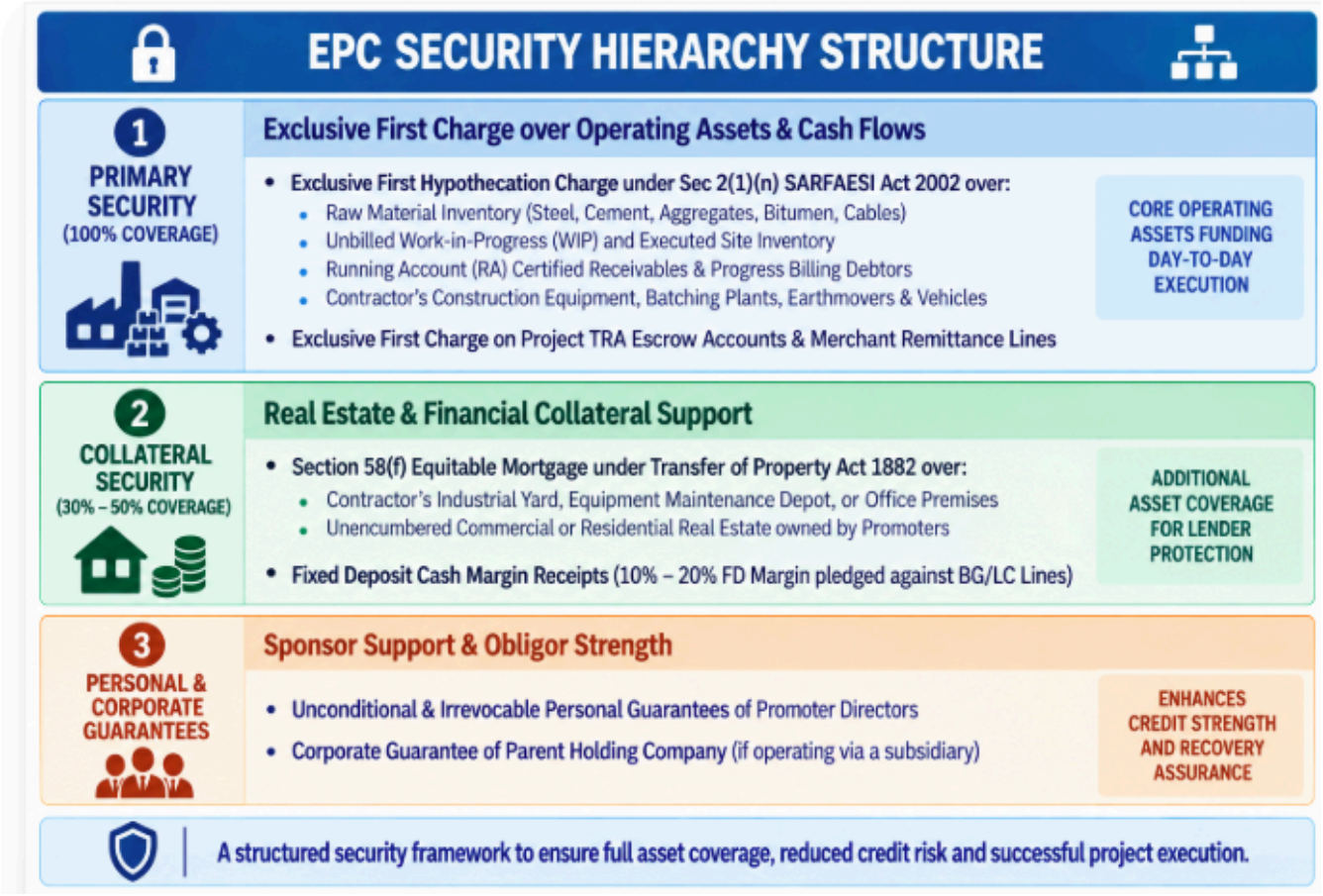


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Module 6: Security Architecture, Documentation & Legal Covenants

6.1 Security Hierarchy & Hypothecation Charges

EPC Funding security structures combine primary project charges with collateral asset mortgages:



6.2 Statutory Filings & Perfection of Security

To establish priority charge enforcement across Indian credit registers, lenders mandate strict statutory filings:

1. **ROC Form CHG-1 Filing (Within 30 Days):** Under **Section 77 of the Companies Act, 2013**, every charge created on the contractor's assets, inventory, receivables, or bank accounts must be registered with the Registrar of Companies (ROC) within 30 days. Failure to file CHG-1 renders the charge void against liquidators in insolvency.

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2. **CERSAI Portal Registration:** In compliance with the **SARFAESI Act, 2002**, all security interests created over movable inventory, equipment, receivables, and immovable property mortgages must be registered on the Central Registry of Securitisation Asset Reconstruction and Security Interest (CERSAI) portal to prevent double-pledging.
3. **National E-Governance Services Limited (NeSL) Filings:** Financial debt contracts and default records are submitted to NeSL Information Utility (IU) to establish legal proof of debt in Section 7 Insolvency and Bankruptcy Code (IBC) proceedings.

6.3 Key Legal Covenants

- **Employer Direct Payment Mandate:** Irrevocable agreement executed by the employer agreeing to remit all RA bill payments directly into the TRA Escrow Account.
- **Negative Lien on Project Assets:** Restricts the contractor from creating secondary charges or selling construction equipment without prior written lender consent.
- **Minimum Order Book Maintenance:** Mandates that the contractor maintain an unexecuted order book equal to at least 2.5x annual turnover throughout the facility tenure.

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Module 7: Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields

7.1 Indicative Tenure Structure

- **Fund-Based Facilities (CC / WCDL / Bill Discounting):** 12-month revolving annual facility, subject to annual credit renewal, financial review, and Drawing Power (DP) audits. Individual WCDL tranches and RA Bill Discounting drawings carry tenures of 30 to 180 days matching the client payment cycle.
- **Non-Fund Based Guarantee Facilities (PBG / MABG / Tender BGs):** Issued on a transaction-specific basis. Bid Bonds carry tenures of 90 to 180 days. Mobilization Advance BGs carry tenures of 12 to 36 months (amortized as work completes). Performance BGs carry tenures matching the construction period plus the 12-to-24 month Defect Liability Period (total tenure 2 to 6 years).

7.2 Commercial Pricing Mechanics & Tax Shields

Financial pricing for EPC Umbrella Lines combines floating interest spreads on fund-based drawings with commission rates on non-fund-based guarantees:

Fund-Based Borrowing Cost (CC/WCDL)=External Benchmark (EBLR) / MCLR+Risk Spread (1.50%to 3.50%p.a.)

$$Non\ -\ Fund\ Based\ BG\ Commission\ Rate\ =\ 0.75\%\ to\ 2.25\%\ p.a.\ on\ Face\ Value$$

Income Tax Act Section 36(1)(iii) Interest Tax Shield

Under **Section 36(1)(iii) of the Income Tax Act, 1961**, all interest paid on fund-based working capital borrowing (CC, WCDL, Bill Discounting) and all commission fees paid for Bank Guarantees and Letters of Credit are 100% tax-deductible as business operating expenses.

At the standard Indian corporate tax rate of **25.168%** (22% base tax + 10% surcharge + 4% cess under Section 115BAA), the effective post-tax cost of EPC funding is calculated as:

$$Effective\ Post\ -\ Tax\ Borrowing\ Rate\left(K_{d,post}\right) = Pre\ -\ Tax\ Nominal\ Rate\left(K_{d}\right) \times \left(1\ -\ T_{corp}\right)$$

- **For a 9.25% p.a. Pre-Tax Fund-Based Interest Rate:**

$$K_{d,post} = 9.25\% \times (1 - 0.25168) = 6.922\%\ p.a.$$

- **For a 1.25% p.a. Pre-Tax Non-Fund Based BG Commission Rate:**

$$Effective\ Post\ -\ Tax\ BG\ Fee\ =\ 1.25\% \times (1\ -\ 0.25168) = 0.935\%\ p.a.$$

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Additionally, under **Section 43B of the Income Tax Act**, interest deductions are allowed on an actual paid basis, rewarding disciplined debt servicing.

Ind AS 115 Revenue Accounting (Percentage of Completion Method - POCM)

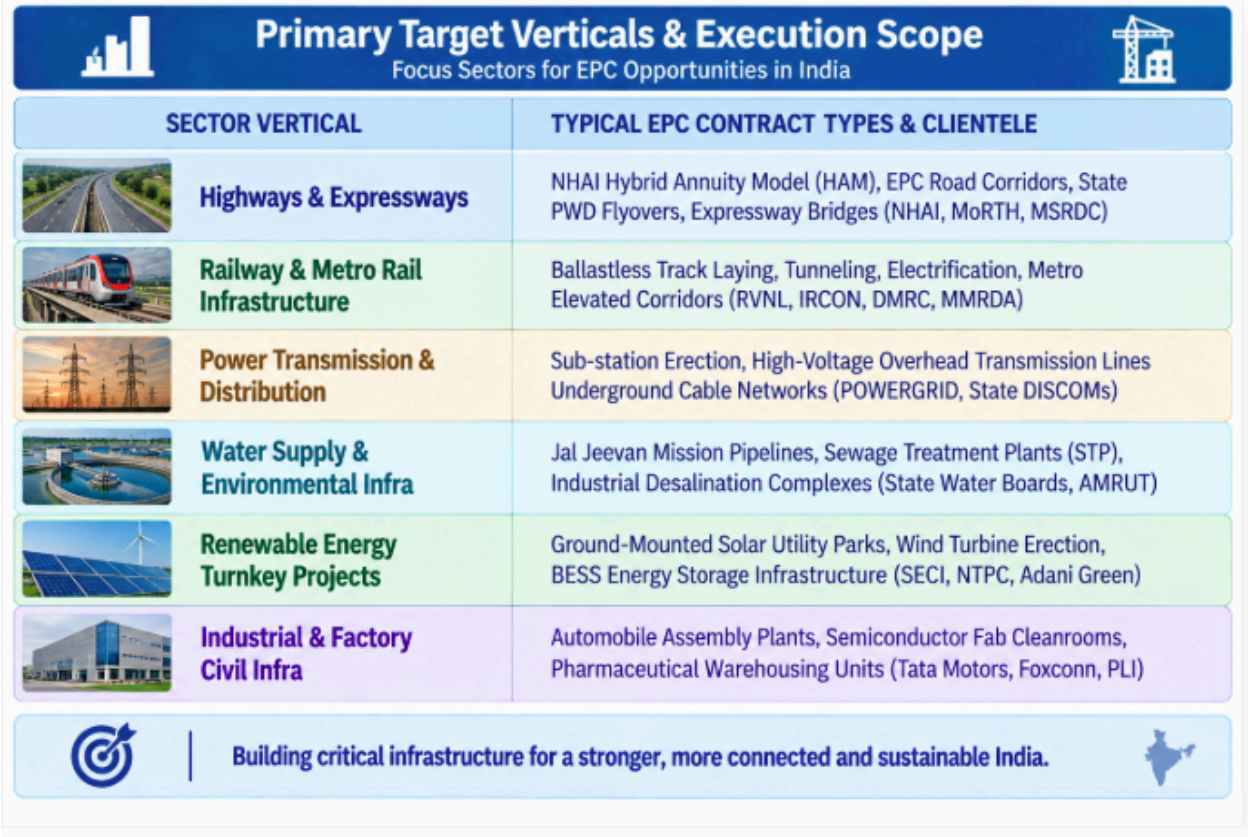
Under **Ind AS 115 (Revenue from Contracts with Customers)**, EPC contractors recognize revenue over time using the **Input Method** (Percentage of Completion Method - POCM). Revenue recognized reflects cumulative contract costs incurred relative to total estimated contract costs. Interest expenses on fund-based working capital are expensed directly to the P&L as finance costs, optimizing operating profit margins.

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Module 8: Suitable Industries, Business Profiles & Market Alignment

8.1 Primary Target Sectors

EPC Funding is specifically structured for contractors operating across high-growth turnkey infrastructure and industrial sectors:



8.2 Market Footprint & Commercial Credit Snapshot

According to commercial credit metrics published across Indian banking and rating agency studies (CRISIL, ICRA, SIDBI MSME Pulse), **Construction Contractors, Infrastructure Builders, and Capital Goods Manufacturers account for over ₹12 Lakh Crores in active commercial credit** across Indian banks and NBFCs.

Furthermore, state-owned commercial trade platforms such as State Bank of India's **e-DFS (Electronic Dealer Financing Scheme, at ₹49,261 Crores)** and **e-VFS (Electronic Vendor Financing Scheme, at ₹28,092 Crores)** highlight the rapid digitization of vendor and contractor supply chain financing across India's infrastructure ecosystem.

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Module 9: Key Advantages and Strategic Limitations

9.1 Key Advantages

1. **Multiplies Order Book Capacity:** Unlocks Bank Guarantees and Letters of Credit without requiring 100% cash margins, allowing contractors to bid for contracts up to 10x their net worth.
2. **Mitigates 90-Day Billing Certification Lags:** RA Bill Discounting provides immediate cash liquidity, preventing site work stoppages during client administrative delays.
3. **Unlocks Interest-Free Client Capital:** Mobilization Advance BGs allow contractors to receive 10%–15% advance funding directly from employers at zero interest cost.
4. **Optimizes Post-Tax Borrowing Costs:** Section 36(1)(iii) tax shields reduce effective borrowing costs to 6.92% p.a.
5. **Protect Promoters' Equity Control:** Non-dilutive debt financing preserves 100% equity ownership for company promoters.

9.2 Strategic Limitations & Risk Factors

1. **Client Payment Delay & Dispute Risk:** If an employer disputes RA bill measurements or delays payments beyond 90 days, drawing power drops, creating cash deficits.
2. **Raw Material Price Volatility:** Unhedged spikes in steel, cement, or bitumen prices can erode profit margins on fixed-price LSTK contracts lacking price variation clauses (PVC).
3. **On-Demand Bank Guarantee Invocation Risk:** Performance BGs in India are legally unconditional instruments. Unjustified invocation by employers forces immediate bank payment, creating an instant debt default for the contractor.
4. **Strict CERSAI & ROC Compliance Requirements:** Failure to register ROC Form CHG-1 within 30 days invalidates charges against liquidators in insolvency.
5. **RBI IRACP 90-Day Asset Classification Rules:** Overdue cash credit drawings exceeding 90 days result in immediate Special Mention Account (SMA-2) and Non-Performing Asset (NPA) classification.

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Module 10: Comprehensive Real-World Case Study (Indian Rupee Models)

10.1 Case Background: Mumbai EPC Construction & Engineering Pvt. Ltd. (MECEPL)

- **Company Profile:** Mid-market infrastructure EPC contractor headquartered in Mumbai, specializing in highway expansion, elevated flyovers, and bridge structures for NHAI and State PWDs.
- **Track Record:** 14 years of execution history; net worth of ₹80.00 Crores; CRISIL BBB/Stable credit rating; CIBIL Commercial Rank CMR 3.
- **New Awarded Contract:** Awarded a **₹300.00 Crore Highway & Flyover EPC Contract** by NHAI under an Item-Rate / BOQ structure with a 24-month completion deadline and a 12-month Defect Liability Period (DLP).
- **Contractual Mandates:**
  - Performance Bank Guarantee (PBG): 5% of contract value = **₹15.00 Crores** (Tenure = 36 Months).
  - Mobilization Advance BG (MABG): 10% of contract value = **₹30.00 Crores** (Tenure = 24 Months).
  - Initial Site Mobilization Outlay: **₹15.00 Crores** (Equipment setup, labor camps, asphalt plant).
  - Peak Unbilled WIP & RA Bill Certification Lag: **₹30.00 Crores** (90-day billing cycle).

10.2 Strategic Funding Options Evaluated

MECEPL evaluated two distinct capital sourcing pathways to execute the ₹300.00 Crore NHAI contract:

FINANCING PATHWAY COMPARISON

Two Approaches. A Clear Advantage for Structured EPC Funding.

PARAMETER	PATHWAY A: UNSECURED HIGH-COST DEBT & HIGH FD MARGIN	PATHWAY B: STRUCTURED BANK EPC FUNDING UMBRELLA FACILITY
Total Credit Limits Sized	₹45.00 Cr BGs + ₹15.00 Cr Unsecured Business Loans	₹90.00 Cr Total Umbrella Line: - ₹30.00 Cr FB CC/WCDL Line - ₹45.00 Cr NFB BG Line (PBG/MABG) - ₹15.00 Cr NFB Inland Usance LC
Bank Guarantee Cash FD Margin (Pledged against PBG & MABG)	25.0% Cash Margin FD Required (₹11.25 Crores)	10.0% Cash Margin FD Required (₹4.50 Crores)
Net Promoter Equity Cash Preserved	INR 0.00 (₹11.25 Cr locked in FDs)	+₹6.75 Crores Preserved Liquid Cash Equity for New Bids
Fund-Based Interest Rate	16.50% p.a. Unsecured Business Loans	9.25% p.a. EBLR-linked Cash Credit / WCDL
NFB BG & LC Commission Fee Rate	2.50% p.a. High Non-Rated Commission Rate	1.25% p.a. Rating-Concessional Commission Fee Rate
Escrow TRA Ring-Fencing	None (Direct Company Acc)	6-Tier TRA Escrow Account with NHAI Payment Mandate

STRUCTURED EPC FUNDING UNLOCKS GREATER CAPACITY, LOWER COSTS AND HIGHER FINANCIAL FLEXIBILITY.

Optimized capital structure today. Stronger project execution tomorrow.

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10.3 Comprehensive 2-Year Mathematical Analysis (INR Crores)

1. Upfront Promoter Cash Equity Preservation Analysis

- **Pathway A (High-Margin 25% FD Requirement):**
  - PBG (₹15.00 Cr) + MABG (₹30.00 Cr) = ₹45.00 Crores BG Exposure.
  - Required 25% FD Cash Margin = $0.25 \times 45.00 = ₹11.25\ Crores$.
- **Pathway B (Structured Bank 10% FD Requirement):**
  - Required 10% FD Cash Margin = $0.10 \times 45.00 = ₹4.50\ Crores$.
  - **Net Promoter Cash Equity Preserved:** $11.25 - 4.50 =+ ₹6.75\ CRORES$.

2. Fund-Based Interest & Tax Shield Calculations (2-Year Horizon)

- **Pathway A (Unsecured Borrowing @ 16.50% p.a. on ₹30.00 Cr Average Utilization):**
  - Gross Pre-Tax Annual Interest = $30.00 \times 16.50\% = ₹4.95\ Crores/year$.
  - 2-Year Gross Interest Outflow = $4.95 \times 2 = ₹9.90\ Crores$.
  - Less Section 36(1)(iii) Tax Shield (@ 25.168%):
    $9.90 \times 25.168\% = ₹2.49\ Crores$.
  - **Pathway A Net 2-Year Post-Tax Interest Cost:** $9.90 - 2.49 = ₹7.41\ Crores$.
- **Pathway B (Structured Bank Line @ 9.25% p.a. on ₹30.00 Cr Average Utilization):**
  - Gross Pre-Tax Annual Interest = $30.00 \times 9.25\% = ₹2.775\ Crores/year$.
  - 2-Year Gross Interest Outflow = $2.775 \times 2 = ₹5.55\ Crores$.
  - Less Section 36(1)(iii) Tax Shield (@ 25.168%):
    $5.55 \times 25.168\% = ₹1.40\ Crores$.
  - **Pathway B Net 2-Year Post-Tax Interest Cost:** $5.55 - 1.40 = ₹4.15\ Crores$.
  - **Direct Post-Tax Fund-Based Savings:** $7.41 - 4.15 =+ ₹3.26\ CRORES$.

3. Non-Fund Based Commission Fee Analysis (2-Year Horizon)

- **Pathway A (2.50% Commission Rate on ₹45.00 Cr BG Line):**
  - Pre-Tax Annual Commission = $45.00 \times 2.50\% = ₹1.125\ Crores/year$.
  - 2-Year Gross Commission Outflow = $1.125 \times 2 = ₹2.25\ Crores$.
  - Less Section 36(1)(iii) Tax Shield (@ 25.168%):
    $2.25 \times 25.168\% = ₹0.57\ Crores$.

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- **Pathway A Net Post-Tax Commission Cost:** $2.25 - 0.57 = ₹1.68\ Crores$.

- **Pathway B (1.25% Commission Rate on ₹45.00 Cr BG Line):**
  - Pre-Tax Annual Commission = $45.00 \times 1.25\% = ₹0.5625\ Crores/year$.
  - 2-Year Gross Commission Outflow = $0.5625 \times 2 = ₹1.125\ Crores$.
  - Less Section 36(1)(iii) Tax Shield (@ 25.168%): $1.125 \times 25.168\% = ₹0.283\ Crores$.
  - **Pathway B Net Post-Tax Commission Cost:** $1.125 - 0.283 = ₹0.842\ Crores$.
  - **Direct Post-Tax Non-Fund Based Savings:** $1.68 - 0.842 =+ ₹0.838\ CRORES$.

4. Opportunity Income Earned on Reinvested Preserved Equity

By selecting Pathway B, MECEPL preserves **₹6.75 Crores in liquid cash equity** (due to lower FD margins). Reinvesting this capital into operational site execution across a second PWD contract yields an estimated **18.00% p.a. Return on Capital Employed (ROCE)**: * 2-Year Gross ROCE Reinvestment Income = $6.75 \times 18.00\% \times 2 = ₹2.43\ Crores$. * Less Corporate Income Tax (@ 25.168%): $2.43 \times 25.168\% = ₹0.61\ Crores$. * **Net Post-Tax Reinvestment Income Earned:** $2.43 - 0.61 =+ ₹1.82\ CRORES$.

5. Total Opportunity-Adjusted Net Financial Wealth Surplus

Combining all direct savings and reinvestment returns:

Total Net Wealth Surplus=FB Savings (₹3.26 Cr)+NFB Savings (₹0.84 Cr)+Reinvestment Income (₹1.82 Cr)−Bank Setup Fee (₹0.15 Cr)

Total Net Wealth Surplus=3.26+0.84+1.82−0.15=+₹5.77 CRORES (+INR 5,77,00,000.00)

Furthermore, MECEPL's project Debt Service Coverage Ratio (DSCR) expanded from a tight **1.18x** under Pathway A to a prime **2.05x** under Pathway B!

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Module 11: Professional Certification Examination & Detailed Rationales

This module contains 10 scenario-based examination questions designed to test complete mastery of EPC Funding principles, statutory regulations, coverage ratios, and tax mechanics.

Question 1

An EPC contractor awarded a ₹200 Crore highway contract receives a 10% Mobilization Advance from NHAI. The employer requires an equivalent Mobilization Advance BG (MABG). The contractor's bank mandates a 15% FD cash margin. What is the net cash liquidity unlocked by the contractor via the mobilization advance? A) ₹20.00 Crores
B) ₹17.00 Crores
C) ₹3.00 Crores
D) ₹15.00 Crores

Answer: **B) ₹17.00 Crores**
Rationale: Mobilization Advance = $10\% \times$ ₹$200\ Cr = ₹20.00\ Crores$. MABG issued = ₹20.00 Crores. FD margin required = $15\% \times$ ₹$20.00\ Cr =$ ₹$3.00\ Crores$. Net cash unlocked = ₹$20.00\ Cr -$ ₹$3.00\ Cr =$ ₹$17.00\ Crores$.

Question 2

Under the Tandon Committee Second Method of Lending (MPBF), an EPC contractor has Total Current Assets of ₹100 Crores and Other Current Liabilities (excluding bank borrowing) of ₹30 Crores. What is the Maximum Permissible Bank Finance (MPBF)? A) ₹70.00 Crores
B) ₹50.00 Crores
C) ₹45.00 Crores
D) ₹25.00 Crores
Answer: **C) ₹45.00 Crores**
Rationale: Working Capital Gap (WCG) = $TCA - OCL = 100 - 30 =$ ₹$70.00\ Crores$.
Minimum Stipulated NWC = $0.25 \times TCA = 0.25 \times 100 =$ ₹$25.00\ Crores$.
$MPBF = WCG - Minimum\ NWC = 70 - 25 =$ ₹$45.00\ Crores$ (or $0.75 \times 100 - 30 = 45.00\ Crores$).

Question 3

Which statutory provision governs the mandatory 30-day registration of a hypothecation charge created on an EPC contractor's machinery, WIP, and receivables with the Registrar of Companies? A) Section 58(f) of the Transfer of Property Act, 1882

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B) Section 77 of the Companies Act, 2013
C) Section 138 of the Negotiable Instruments Act, 1881
D) Section 25 of the Payment and Settlement Systems Act, 2007
Answer: **B) Section 77 of the Companies Act, 2013**
Rationale: Section 77 of the Companies Act, 2013 mandates that every company creating a charge on its property or assets (tangible or intangible) must register ROC Form CHG-1 within 30 days of creation.

Question 4

An EPC company incurs ₹10.00 Crores in pre-tax interest on its Cash Credit and WCDL facilities. Assuming the corporate tax rate is 25.168% under Section 115BAA of the Income Tax Act, 1961, what is the net post-tax interest cost after applying the Section 36(1)(iii) tax shield? A) ₹10.00 Crores
B) ₹7.483 Crores
C) ₹2.517 Crores
D) ₹5.000 Crores
Answer: **B) ₹7.483 Crores**
Rationale: Section 36(1)(iii) allows interest paid as a 100% tax-deductible expense. Tax savings = $10.00 \times 25.168\% = ₹2.5168\ Crores$. Net post-tax cost = $10.00 - 2.5168 = ₹7.4832\ Crores$.

Question 5

Under Tandon Committee Second Method rules, what is the minimum required Current Ratio (Total Current Assets / Total Current Liabilities) for an EPC contractor seeking a working capital line exceeding ₹5 Crores? A) 1.00:1
B) 1.15:1
C) 1.33:1
D) 2.00:1
Answer: **C) 1.33:1**
Rationale: The Tandon Committee Second Method mandates that at least 25% of gross current assets must be funded via net working capital (long-term funds), establishing a mathematical minimum Current Ratio of $100/(100 - 25) = 100/75 = 1.33:1$.

Question 6

What is the primary difference between a Performance Bank Guarantee (PBG) and a Mobilization Advance Bank Guarantee (MABG)? A) PBGs fund raw material purchases, while MABGs fund machinery leasing.
B) PBGs secure the employer against defective work/non-performance, while MABGs secure the

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employer's cash advance granted for site setup.
C) PBGs carry a 100% cash margin requirement, while MABGs require zero cash margin.
D) PBGs are issued under RBI guidelines, while MABGs are governed by SEBI.
Answer: **B) PBGs secure the employer against defective work/non-performance, while MABGs secure the employer's cash advance granted for site setup.**
Rationale: PBGs guarantee that the contractor will execute the project per technical specifications (5%–10% value). MABGs secure the 10%–15% interest-free cash advance disbursed by the employer during mobilization.

Question 7

Which accounting standard governs revenue recognition for EPC turnkey contractors using the Input Method (Percentage of Completion Method - POCM)? A) Ind AS 16
B) Ind AS 23
C) Ind AS 115
D) Ind AS 109
Answer: **C) Ind AS 115**
Rationale: Ind AS 115 (Revenue from Contracts with Customers) governs revenue recognition over time based on performance obligations satisfied, utilizing the Input Method (POCM) to align recognized revenue with cumulative contract costs.

Question 8

An EPC contractor's RA bill of ₹10.00 Crores is certified by the client. The contract stipulates a 10% Mobilization Advance recovery and a 5% Retention Money deduction. What is the net cash remitted into the contractor's TRA Escrow Account? A) ₹10.00 Crores
B) ₹9.00 Crores
C) ₹8.50 Crores
D) ₹7.50 Crores
Answer: **C) ₹8.50 Crores**
Rationale: Gross RA Bill = ₹10.00 Crores. Less Mobilization Advance recovery ($10\% = ₹1.00\ Cr$). Less Retention Money deduction ($5\% = ₹0.50\ Cr$). Net cash remitted = $10.00 - 1.00 - 0.50 = ₹8.50\ Crores$.

Question 9

What legal recourse allows an EPC contractor's lender to enforce security over pledged machinery and inventory without filing a prior court lawsuit? A) Arbitration and Conciliation Act, 1996
B) Section 13 of the SARFAESI Act, 2002
C) Negotiable Instruments Act, 1881

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D) Insolvency and Bankruptcy Code, 2016
Answer: **B) Section 13 of the SARFAESI Act, 2002**
Rationale: Section 13 of the SARFAESI Act, 2002 empowers secured creditors to enforce security interests (hypothecated equipment, inventory) without court intervention after issuing a 60-day demand notice following NPA classification.

Question 10

If an EPC contractor's Cash Credit account remains continuously overdrawn beyond the sanctioned limit / Drawing Power for more than 90 days, how is the asset classified under RBI IRACP norms? A) Special Mention Account 0 (SMA-0)
B) Special Mention Account 1 (SMA-1)
C) Non-Performing Asset (NPA) - Substandard Asset
D) Restructured Standard Asset
Answer: **C) Non-Performing Asset (NPA) - Substandard Asset**
Rationale: Under the Reserve Bank of India (RBI) Prudential Norms on Income Recognition, Asset Classification and Provisioning (IRACP), an overdraft or cash credit account that remains continuously "out of order" for over 90 days must be classified as a Non-Performing Asset (NPA).

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